

4-Channel MULTI-MODE AUDIO AMPLIFIER

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Sometimes we need configurable multichannel amplifiers for experiments or for use in offices or schools. Here is a circuit based on TDA1554Q, configured for multi-mode function with four channel outputs.

Circuit and working

The circuit of the simple 4-channel multi-mode audio amplifier is shown in Fig. 1. It has four channels and each channel can provide around 11W over a load of 2-ohms and around 6W over a load of 4-ohms. The circuit will also work with loads of 4-ohms to 16-ohms.

Mode selection. The multi-mode function includes single-channel and bridge-tied load (BTL) modes. These are controlled using switches S1 and S2. In bridge mode, output power can be around 22W on a 4-ohm load

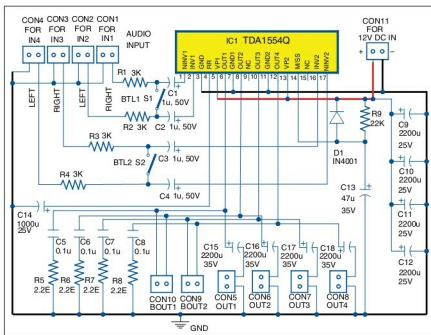


Fig. 1: Circuit of the 4-channel multi-mode audio amplifier

PARTS LIST

Semiconductors:

- IC1 - TDA1554Q audio power amplifier
- D1 - 1N4001 rectifier diode
- D2-D5 - 6A05 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R4 - 3-kilo-ohm
- R5-R8 - 2.2-ohm
- R9 - 22-kilo-ohm
- R10 - 100-ohm

Capacitors:

- C1-C4 - 1 μ F, 50V electrolytic
- C5-C8 - 0.1 μ F ceramic disk
- C9-C12 - 2200 μ F, 25V electrolytic
- C13 - 47 μ F, 35V electrolytic
- C14 - 1000 μ F, 25V electrolytic
- C15-C18, C23 - 2200 μ F, 35V electrolytic
- C19-C22 - 22nF ceramic disk
- C24 - 0.1 μ F, 440V polyester

Miscellaneous:

- X1 - 230V AC primary to 12V, 6A secondary transformer
- F1 - 2.5A fuse
- CON1-CON4 - 3.5mm stereo female panel-mount connector
- CON5-CON10 - 2-way panel-mount spring-loaded speaker connector
- CON11-CON13 - 2-pin connector terminal
- S1-S3 - On/off toggle switch

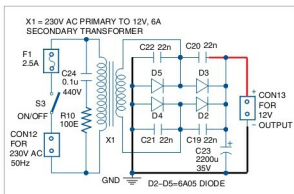


Fig. 2: Power supply circuit of the amplifier

is reduced by two or four times. Consequently, input capacitors C1 through C4 should have enough capacity to give the required sound in the low-frequency range.

Power supply. The circuit for the power supply is shown in Fig. 2. Transformer X1 used during testing was rated at 12V AC, 6A. Ratings

with a good heat-sink, power supply and montage.

Gain of each channel is fixed at 20dB in single-channel mode and 26dB in BTL mode.

Input resistance of each channel of TDA1554Q amplifier (IC1) is around 60-kilo-ohm. When two or four inputs are paralleled, resistance

for X1 can be up to 8A and diodes should preferably be Schotky or other low-drop diodes with 10A rating.

Construction and testing

An actual-size, single-side PCB layout of the 4-channel multi-mode audio amplifier is shown in Fig. 3 and its component layout in Fig. 4.